

Anywhere Avatar: 3D Telepresence with Just a Phone and Laptop

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Abstract

We present *Anywhere Avatar*, a telepresence system that enables full-body and facial avatar reconstruction using a smartphone and a laptop. Users record short videos to generate personalized avatars, which are animated in real time during teleconferencing using webcam-based tracking. Built on pre-trained FLAME and SMPL models, the avatars are rendered in high fidelity using Gaussian splatting. The system runs at near real-time with minimal bandwidth, making expressive 3D telepresence accessible without specialized hardware.

CCS Concepts

• **Computing methodologies** → *Artificial intelligence*; • **Human-centered computing** → *Mixed / augmented reality*;

Keywords

Augmented Reality, Telepresence, Machine Learning, Gaussian Splatting

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1 Introduction and System Features

With the emergence of commercial AR glasses and display technologies, avatar-based immersive 3D teleconferencing such as those

deployed in Apple’s Vision Pro Spatial Persona system [1] is gathering momentum. However, such avatar-based systems have yet to reach the ubiquity of 2D teleconferencing platforms. A primary barrier is the need for expensive tracking setups on the participant side—such as the one featured in Google Beam[6]. As a result, the average consumer equipped only with a laptop and smartphone cannot create or use full-body expressive avatars for 3D meetings.

We redefine this paradigm by enabling detailed full-body and facial avatar reconstruction using only a smartphone and a laptop. A user offline records a short videos capturing their face and body using a phone. A server generates personalized avatars within a day from the captures. Once completed, the avatars can be shared with remote collaborators for teleconferencing.

During a teleconferencing session, the user positions themselves in front of a standard laptop webcam. From the live video stream, our system tracks both body pose and facial expressions, which is communicated to the viewer’s computer.

On the viewer’s side, the tracking data animates both a full-body avatar and a facial avatar, displayed either on a laptop screen or through an XR device. This setup delivers an immersive telepresence experience using devices most users already own.

This functionality is powered by 3D morphable avatar models - pre-trained FLAME [7] and SMPL [8] — which encode human facial expressions and body poses. We train Gaussian splatting models for avatar generation: GaussianBlendshapes [10] for faces and GaussianAvatar [3] for bodies. These models generate 3D Gaussians, rendered into photorealistic video streams displayed to the viewer.

The system operates at near real-time frame rates and requires minimal bandwidth as reported in Section 3. Our system captures the full spectrum of human expressiveness—facial expressions, body gestures, and 3D presence—elements that are often overlooked in traditional teleconferencing. Beyond teleconferencing, our system could also serve as a flexible testbed for future avatar research focusing on various body parts. We hope this work demonstrates

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affordable, expressive, and accessible telepresence, while accelerating the deployment of research-driven avatar technologies to real-world applications.

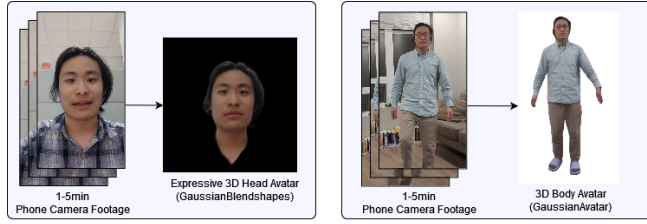


Figure 1: Training Stage of Facial and Body Avatar.

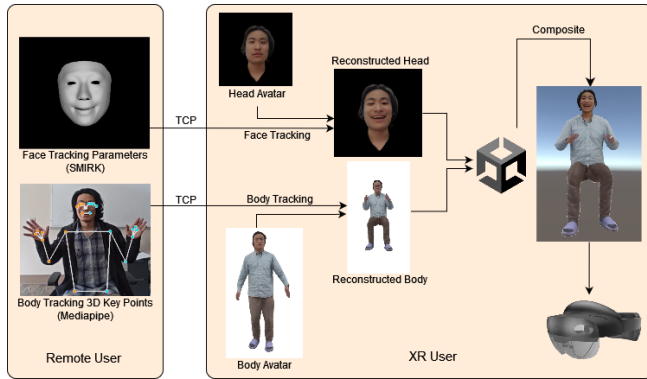


Figure 2: Deployment Stage: Standing Avatars Generalize to Sitting for Meetings

2 System Overview

We present a system that captures, transmits, and reconstructs a remote user's body language and facial expressions in **near real-time**. The remote user generates and transmits body tracking information, which is used to reconstruct and render an avatar into a physical conference room or a digital world using an XR device connected to an XR-ready PC. The system is composed of two main stages: *Training* and *Deployment*.

2.1 Training Stage

Before the meeting, the remote user takes few-minute long video footages of the face making various expressions, as well as the full body at various angles and poses. The videos are sent to a server, where we train a facial avatar as well as a full-body avatar, as illustrated in Figure 1.

Head Avatar. Our system adopts GaussianBlendshapes [10] for the head avatar. Our system first generates the ground-truth FLAME parameters using MICA [12], a metrical FLAME tracker, and segments out the face using the video matting tool provided by MICA. From this we train a gaussian avatar that consists of several static gaussian head blendshapes to be interpolated at run-time.

Body Avatar. Our system adopts GaussianAvatar [3] for body reconstruction. As per InstantAvatar [4], our system invokes OpenPose [2] to generate 2D keypoints of the user's body, Segment Anything Model [5] to segment out the user from the rest of the image, and ROMP [11] to generate the ground-truth SMPL parameters

of the frames. GaussianAvatar then learns a set of 3D Gaussians associated with the SMPL body model that can be driven by parameters. The trained avatars are transferred to the XR user's device in preparation for the deployment stage in the meeting.

2.2 Deployment Stage

During the meeting, the remote user uses a camera to capture body and facial tracking information and sends them over TCP.

Face Tracking. Our system adopts SMIRK and its pre-trained model for facial tracking, thanks to its accuracy as well as its real-time performance. The model generates an estimated camera pose, as well as only 58 FLAME parameters, corresponding to the ones used by GaussianBlendshapes.

Body Tracking. Our system adopts the well-known Mediapipe [9] for body tracking. During run-time, it generates pose keypoints in 3D world space, from which we derive approximate joint rotations, corresponding to the input pose parameters for GaussianAvatar.

As shown in Figure 2, the tracked parameters are sent to the XR user over TCP. The two avatar models perform their respective transforms to generate and render their sets of Gaussians.

Both avatar renderers send their respective frames to shared memory, from which Unity reads their contents and projects them to 3D space.

XR Interface. As shown in Figure 2, the XR user sees the avatar rendered to a 3D world. An AR or MR user will be able to see the remote user placed into the physical environment, such as a conference room, while a VR user will see the remote user in a virtual world.

3 System Performance

Performance Metrics. We evaluate the frame rates and bandwidth of various components of our system.

For the XR user, we benchmark the system with an RTX 4090 Laptop and i9-14900HX, a typical XR-ready setup. Running the facial and body reconstruction together while rendering in a Unity environment, the face renders at 24 FPS (41.0 ms/frame), while the body renders at 12 FPS (82.4 ms/frame).

For the remote user, we benchmark the system with an RTX 4070 Laptop and an Core Ultra 9 185H. The face tracking from SMIRK together with Mediapipe's body tracking reaches 20 FPS (50.0 ms/frame).

Each frame, our system transmits 61 parameters generated by SMIRK (4 bytes each), as well as 33 keypoints from Mediapipe (12 bytes each), for a total of 0.625 KB/frame, or 12.5 KB/s.

Overall, the system achieves near real-time performance while maintaining minimal bandwidth requirements, making it suitable for telepresence applications over standard internet connections.

4 Conclusion

We introduced *Anywhere Avatar*, a system for near real-time full-body and facial avatar telepresence using only a smartphone and a laptop. We hope this work inspires future research toward scalable, real-time avatar systems. This work was supported by NSF grants IIS 2140645, CCF 2217144, CNS 2106592, CNS 2106463, CNS 2032033, CNS 2346528, CNS 2106150, grant SBC UT Dallas 23011912, and NIH grant R01EY030470.

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